ANATOMY OF THE DIAPHRAGM

BY

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Outline

- Intro'
- Functions
- Origins & Insertion
- Openings
- Relations
- Arterial supply
- Venous drainage
- Lymphatic drainage
- Applied Anatomy

Intro

- The **diaphragm** is a **thin, musculotendinous dome-shaped partition** separating the **thoracic** and **abdominal cavities**.

- It is an **important muscle** of **inspiration** and its **position** changes **constantly** with **respiration**.
- The **fibrous, central part** of the **diaphragm**-the **central tendon** is **slightly depressed** by the **heart**. The **more elevated right** and **left parts** of the **diaphragm** form the **right** and **left domes** of the diaphragm. The **right dome**, supported by the **liver** lies at a slightly **higher level** (a little **inferior** to the **nipple** in the **male**) than the **left dome**.

Functions

- It increases **intra-abdominal pressure**
- It helps in **lymphatic return** and **venous return** due to **pressure changes**.
- It assists in **expulsion** (vomiting, defecation, urination, parturition).
- When it contracts, the **dome** of the **diaphragm** is **partially flattened**; the **abdominal contents** are **displaced downwards**, and the **vertical extent** of the **thoracic cavity** is **increased**, it **decreases intrathoracic pressure**. When it **relaxes**, it **rises** and may go **up** to the **level** of the **fourth rib**, **superior** to the **nipple**, in **full expiration**.

Attachments of the diaphragm

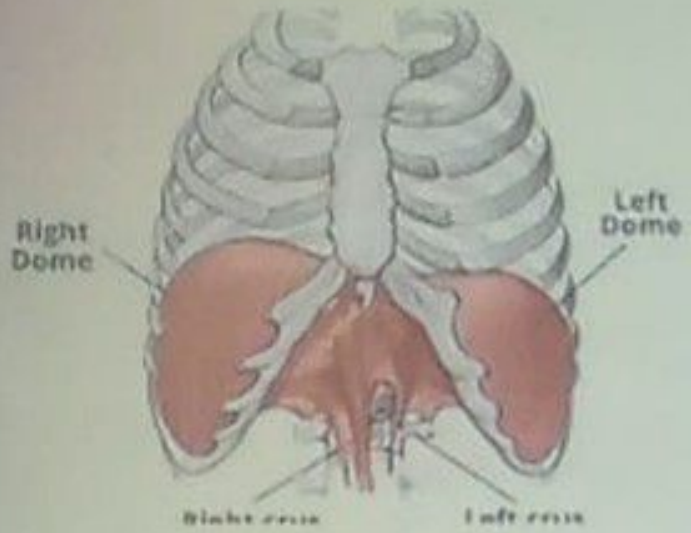
- The **diaphragm** arises from the **margins** of the **inferior aperture** of the **thorax** and **passes** upwards and **inwards** to the **edges** of the **flat, C-shaped central tendon**. The **anterior fibres** run **horizontally backwards**. The **posterior fibres** are **almost vertical** and pass **upwards** to the **central tendon**. There is a **deep recess** between the **posterior thoracic wall** and the **posterior fibres** of the **diaphragm**- the **costodiaphragmatic recess**.
- The **diaphragm** takes **origin** from the **sternum**, the **lower six costal cartilages**, the **upper lumbar vertebra**, and the **three arcuate ligaments**.

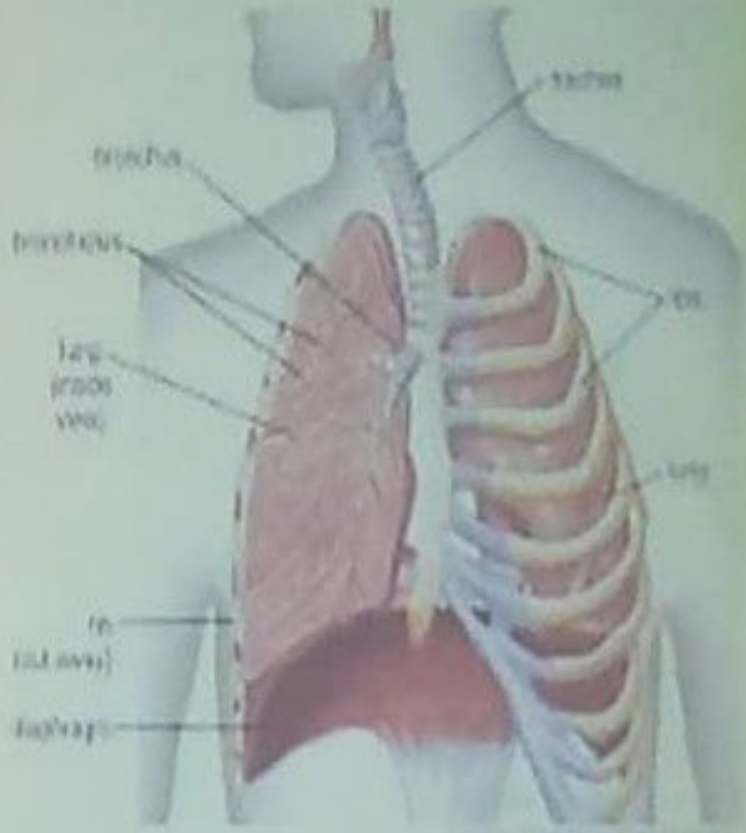
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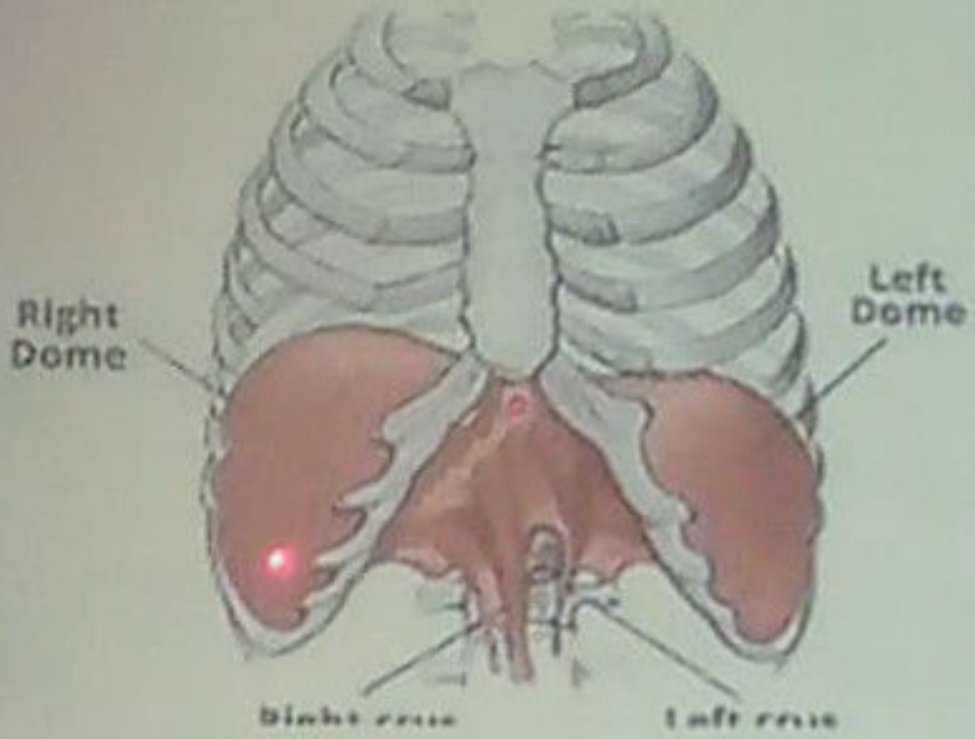
- The **sternal origin** is by **two small slips** which arise from the **xiphoid process** (posterior surface).
- The **costal origin** is by **wide slips** from each of the **lower six costal cartilages**. These **slips** interdigitate with **slips** of **origin** of the **transversus abdominis**.
- The **vertebral (lumbar) origin** is from: (a) the **crura**; and (b) the **arcuate ligaments**.
- The **right** and **left crura** are **thick, fleshy muscle bundles** which are **attached** to the **anterior surfaces** of the **upper lumbar vertebral bodies**. The **left crus** of the **diaphragm** is attached to the **upper two lumbar vertebrae (L1-L2)**.

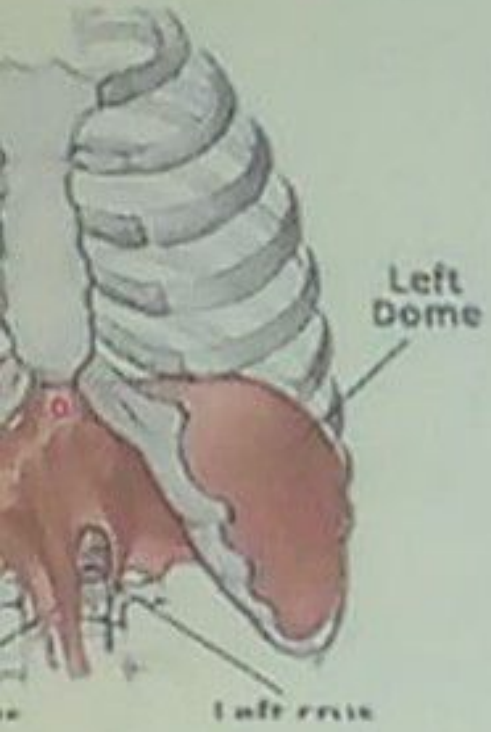
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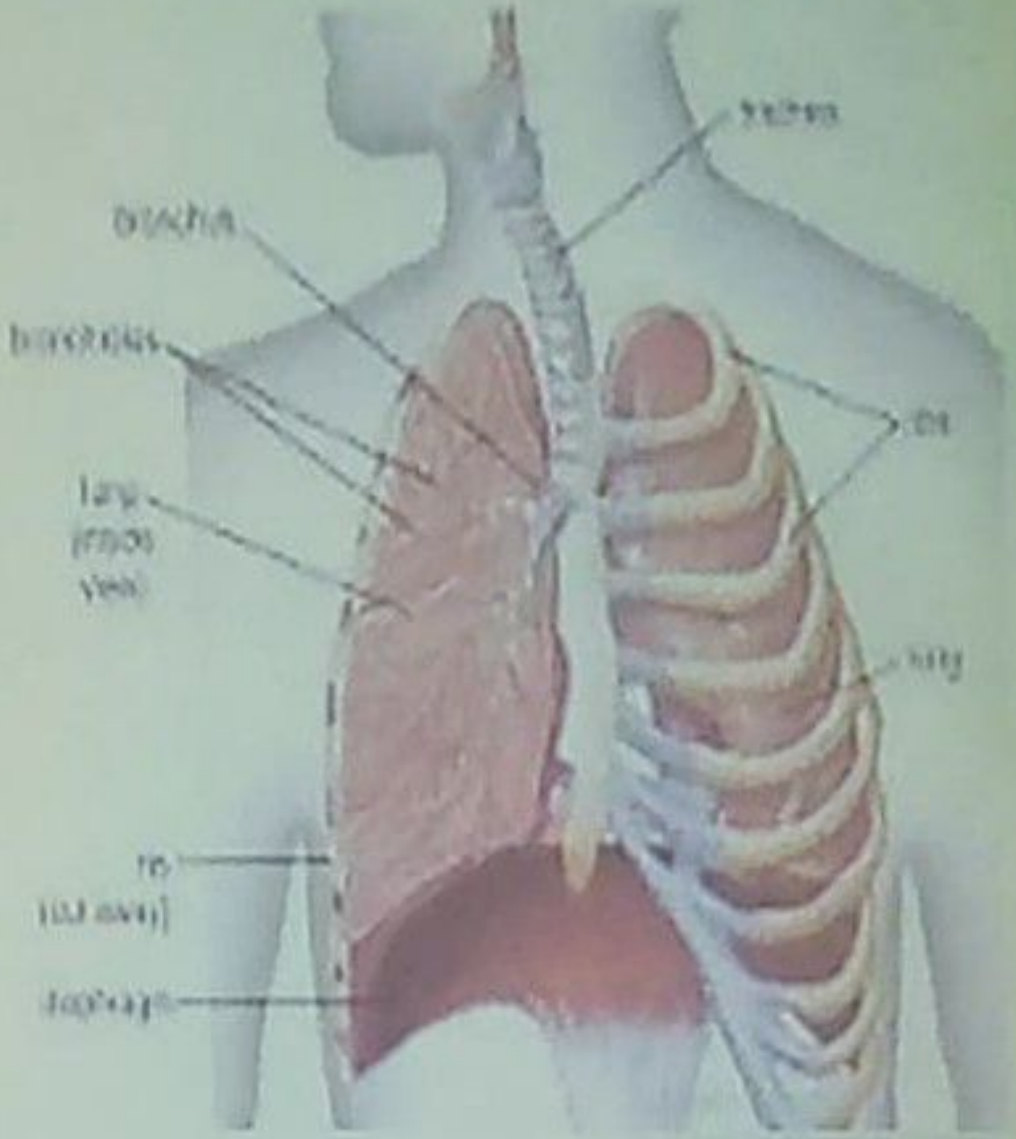
- The **crura** are also **attached** to the **intervening intervertebral discs** and the **tendinous slips** that bridge the **lumbar vessels** on the **vertebral bodies**. They **lie** on **each side** of the **aorta**. As the **fibres** of the **right crus** of the **diaphragm** ascend towards the **central tendon** of the **diaphragm**, they **deviate** to the **left** and encircle the **oesophageal hiatus**. The **right crus** of the **diaphragm** arises from L1-L3.
- The **median arcuate ligament** is a **tendinous band** which unites the **medial sides** of the **right** and **left crura** superiorly. It lies on the **anterior surface** of the **aorta.**
- The **medial arcuate ligament** is a **tendinous thickening** of the **fascia** over the **psoas major**. It stretches from the lateral side of each crus to the transverse process of the first (or second) lumbar vertebra.


Right
Dome
Left
Dome



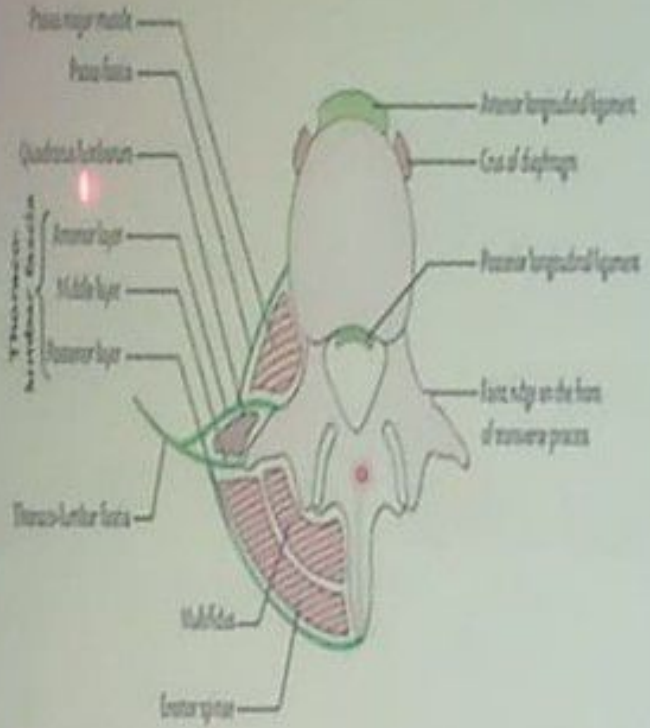

Right
Dome
Left
Dome

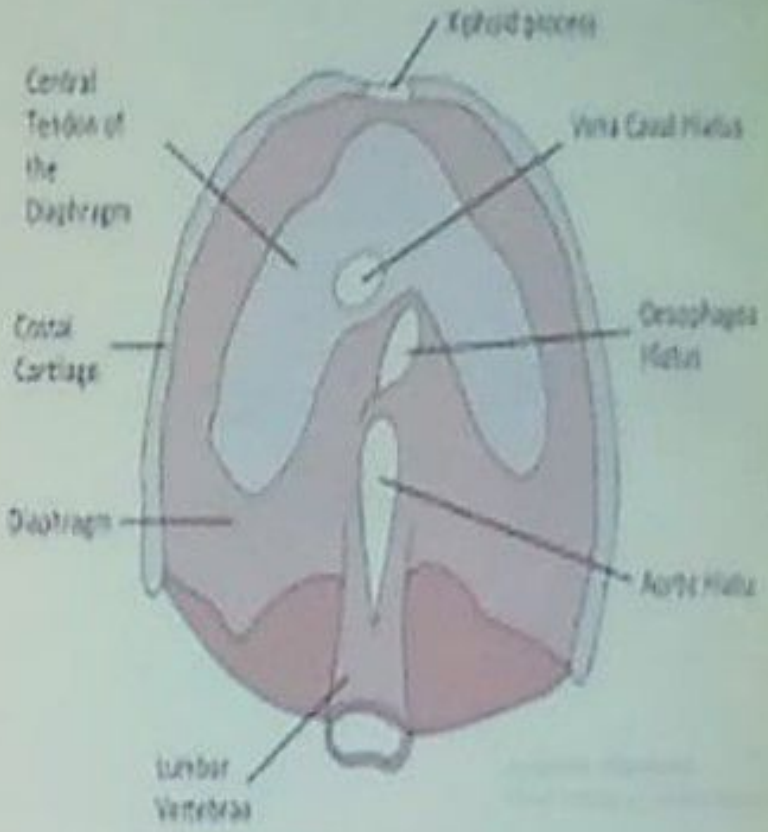

Left
Dome

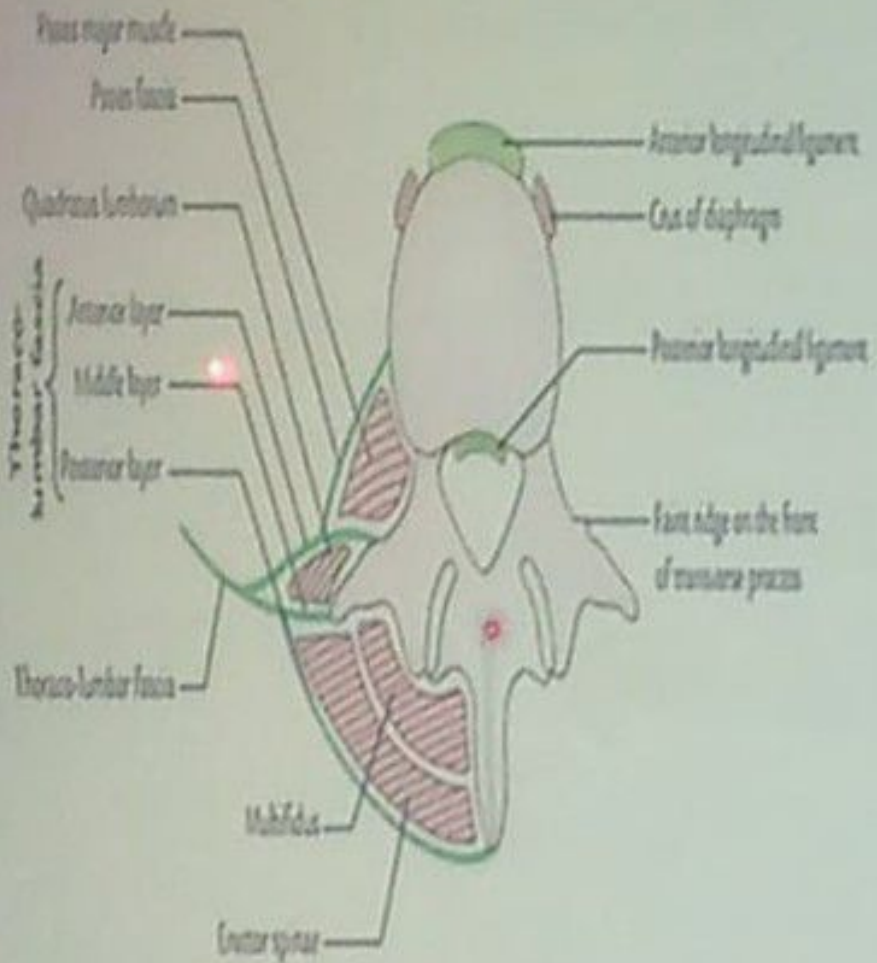


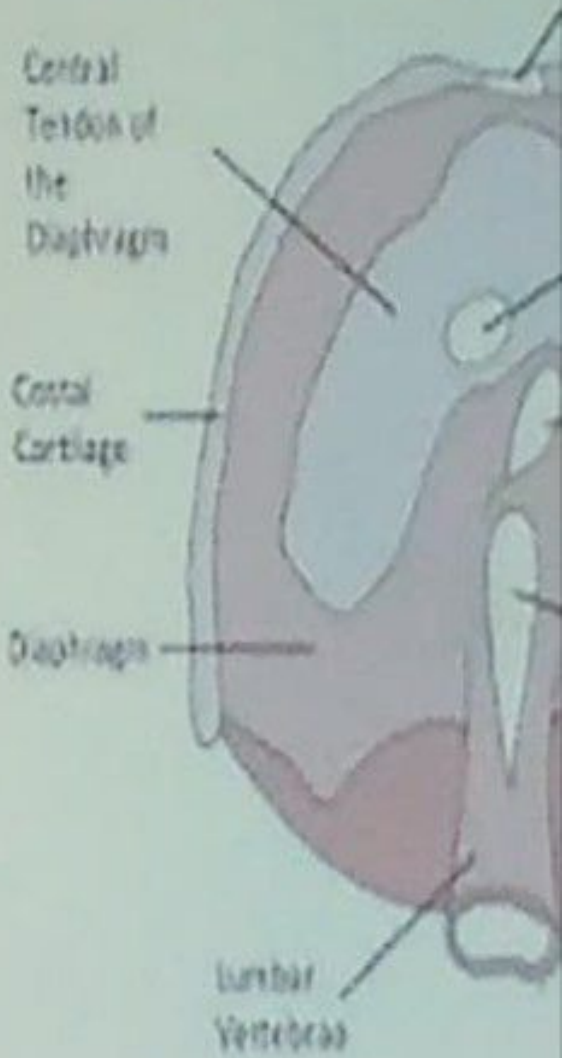
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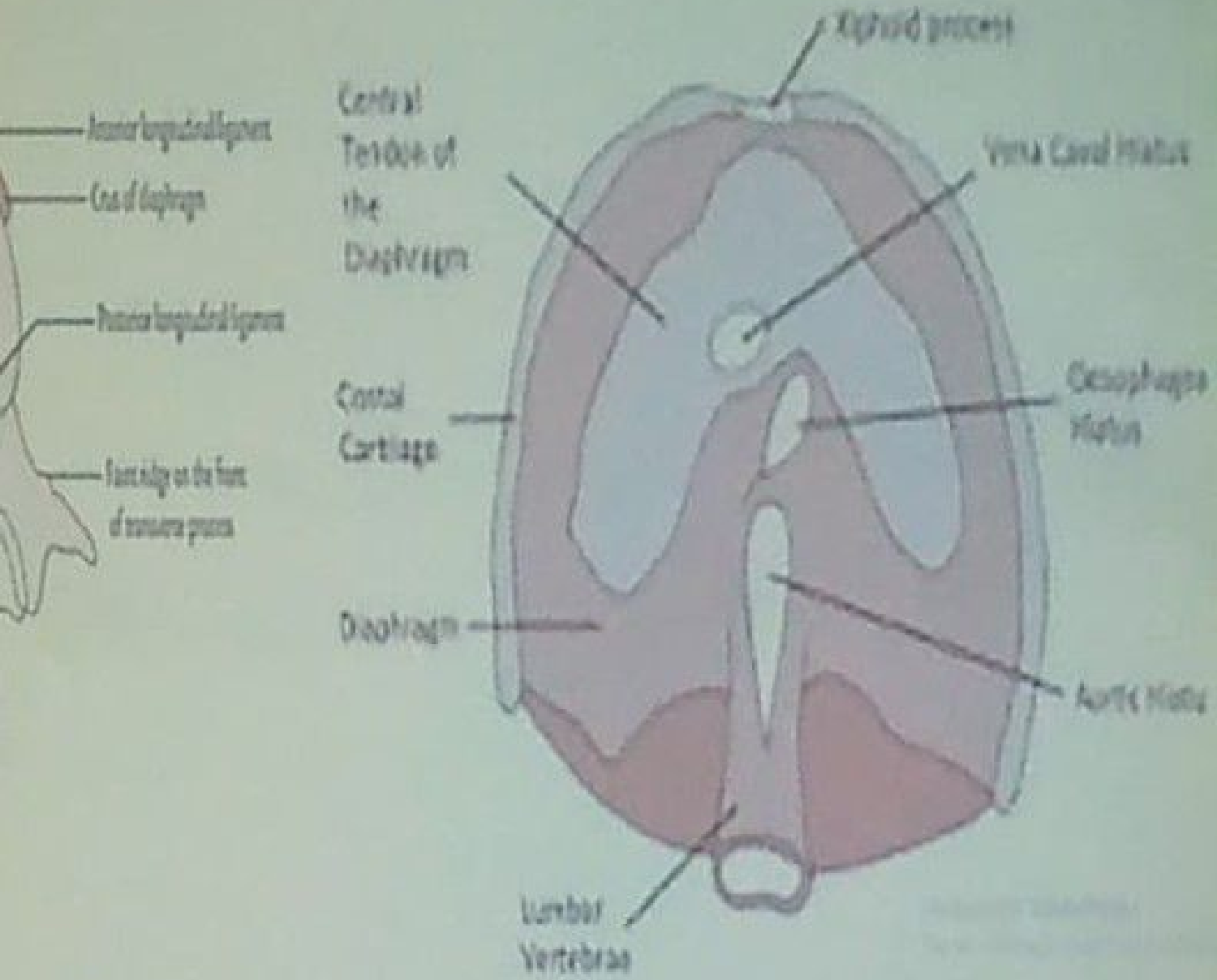
- The **lateral arcuate ligament** is a linear **thickening** of the anterior layer of the thoracolumbar fascia. It extends from the **median arcuate ligament** over the **anterior surface** of the **quadratus lumborum** to the **twelfth rib**.
- ***Fibres*** *from the* ***lateral part*** *of the* ***lateral arcuate ligament*** *may be missing. Such a deficiency is known as the* ***vertebrocostal triangle*** *and is thought to represent an* ***incomplete closure*** *of the* ***embryonic pleuroperitoneal canal****. When the* ***vertebrocostal triangle*** *is present, it* ***leaves*** *the* ***pleura*** *almost* ***directly*** *in* ***contact*** *with the* ***kidney*** *and makes a* ***potential site*** *for* ***herniation*** *of* ***abdominal contents*** *into the* ***thorax****.*


Psoas major muscle
Psoas fascia
Quadratus lumborum
Transversus abdominis muscle
Anterior layer
Middle layer
Posterior layer
Thoraco-lumbar fascia
Anterior longitudinal ligament
Caudal diaphragm
Psoas longitudinal ligament
Lumbar
Erector spinae


Xiphoid process
Central Tendon of the Diaphragm
Vena Caval Hiatus
Costal Cartilage
Oesophageal Hiatus
Diaphragm
Aortic Hiatus
Lumbar Vertebrae

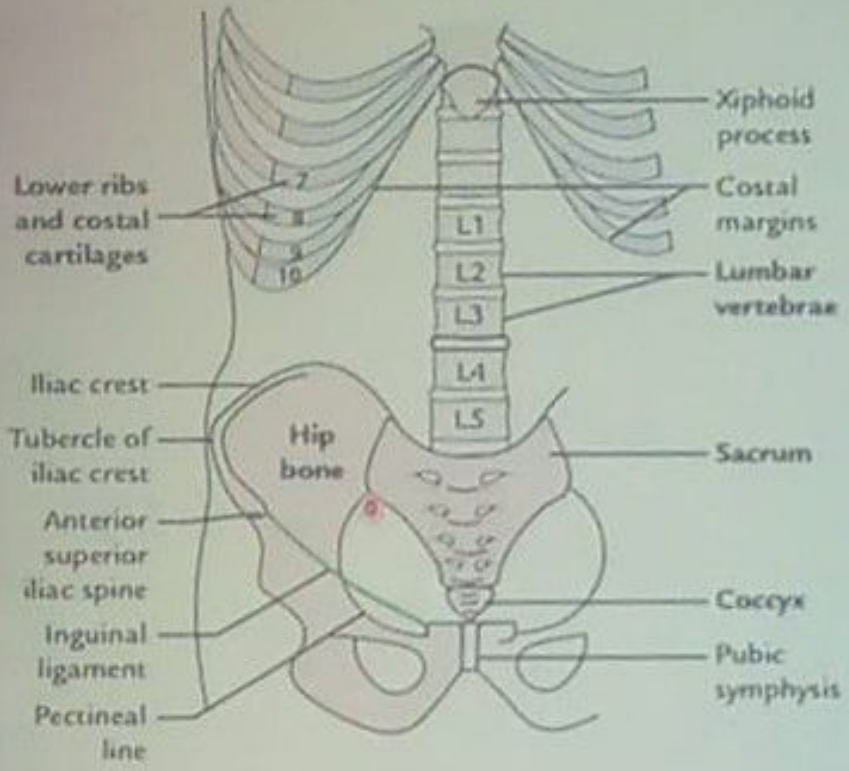

Psoas major muscle
Psoas fascia
Quadratus Lumborum
Anterior layer
Middle layer
Posterior layer
Thoraco-Lumbar fascia
Multifidus
Erector spinae
Anterior longitudinal ligament
Crus of diaphragm
Posterior longitudinal ligament
Fore edge on the fore
of transverse process

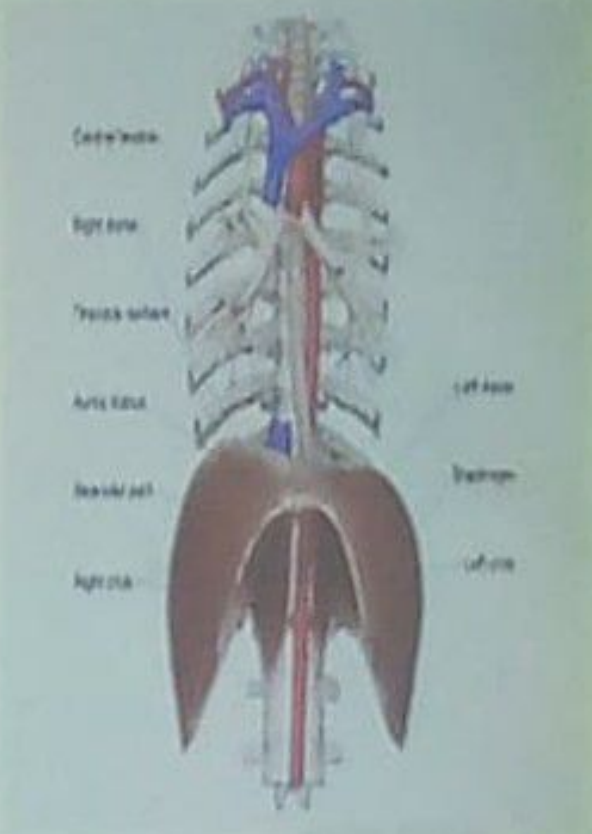

Central
Tendon of
the
Diaphragm
Costal
Cartilage
Diaphragm
Lumbar
Vertebrae

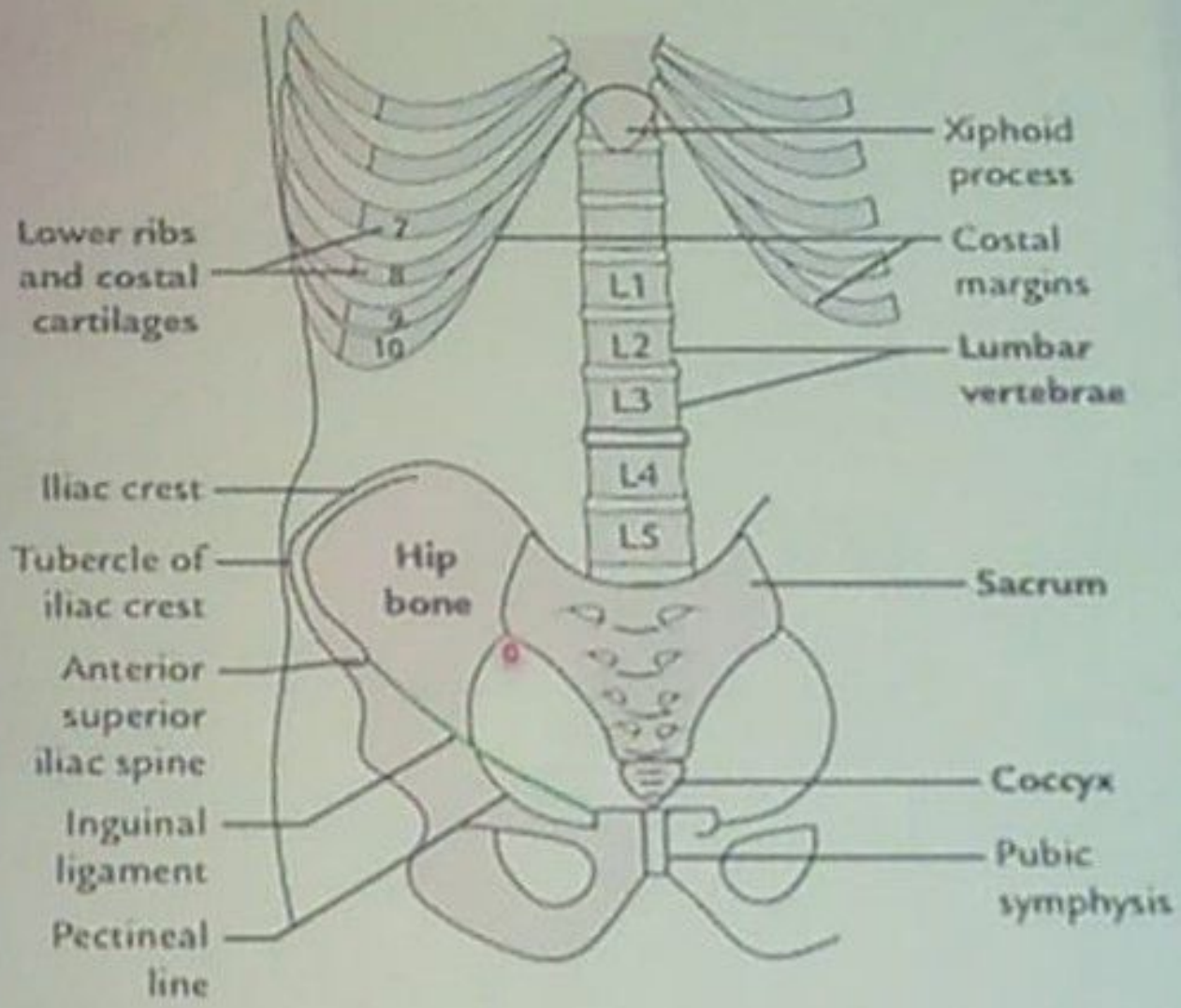

Central
Tendon of
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Vena Caval Hiatus
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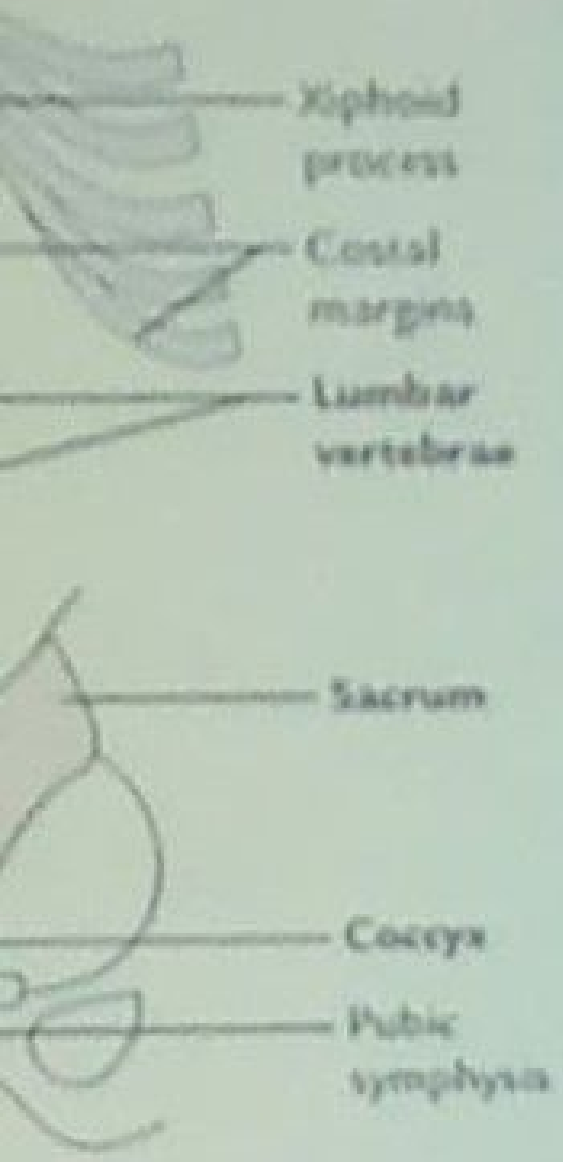
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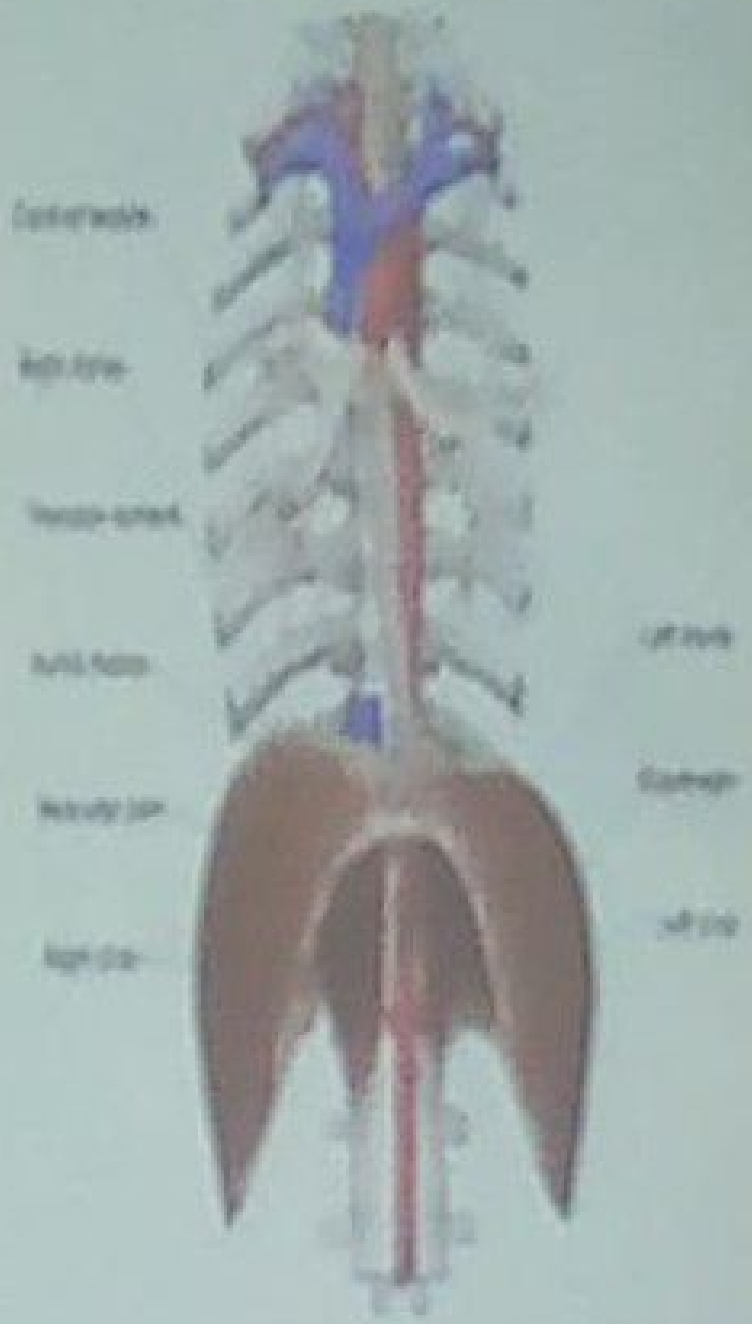
- The **muscle fibres** of the **diaphragm** converge and are **inserted** into the **C-shaped central tendon**. This **tendon** has a **larger median part** which expands **anteriorly** towards the **xiphoid process**, **right** and **left horns** which curve **posteriorly** into the **corresponding halves** of the **diaphragm**.
- **Central tendon** is a strong aponeurotic structure with no bony attachments).


Xiphoid process
Lower ribs and costal cartilages
7
8
9
10
Costal margins
L1
L2
L3
Lumbar vertebrae
L4
L5
Iliac crest
Tubercle of iliac crest
Hip bone
Sacrum
Anterior superior iliac spine
Coccyx
Inguinal ligament
Pubic symphysis
Pectineal line




Xiphoid process
Lower ribs and costal cartilages
7
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Costal margins
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Lumbar vertebrae
Iliac crest
Tubercle of iliac crest
Hip bone
Sacrum
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Coccyx
Inguinal ligament
Pubic symphysis
Pectineal line


Xiphoid process
Costal margin
Lumbar vertebrae
Sacrum
Coccyx
Pubic symphysis



Foramina/Opening in the diaphragm

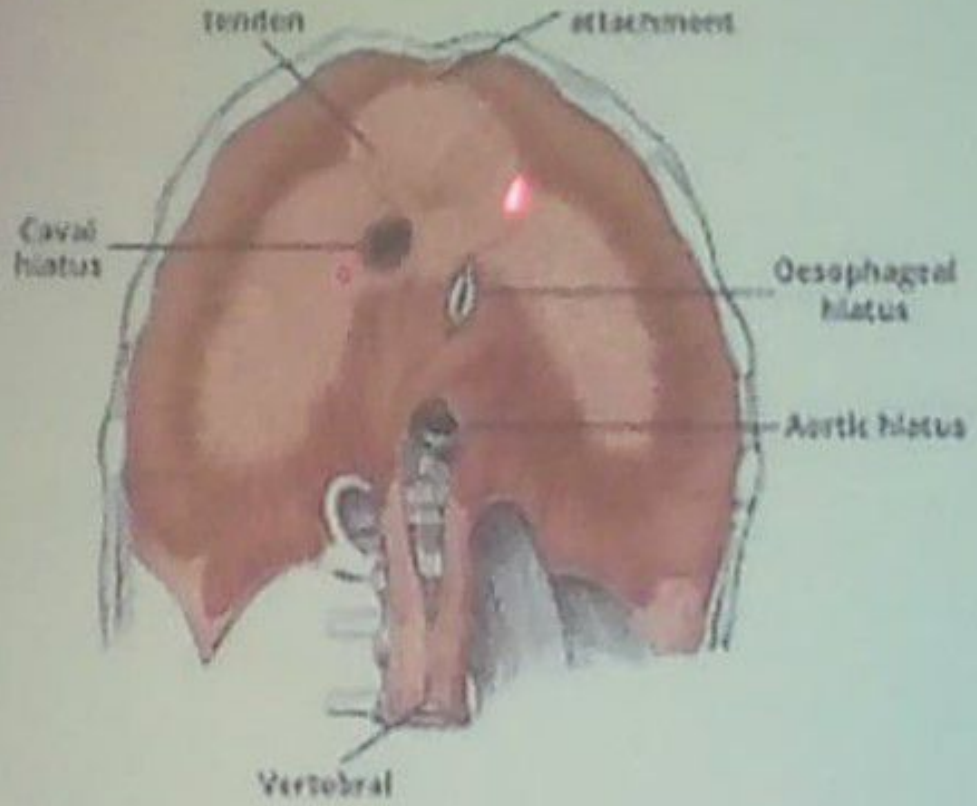
- There are **three large** and **numerous small openings**, or **foramina**, in the **diaphragm**. These **foramina allow** for **passage** of **structures** between the **thoracic** and **abdominal cavities**.
- **Caval hiatus**: The **inferior vena cava** (IVC) **pierces** and **fuses** with the **central tendon** of the **diaphragm**, **2-3 cm** to the **right** of the **median plane** , **approximately** at the **level** of the **eighth thoracic (T8) vertebra**. When the diaphragm **contracts**, it **compresses** the **abdominal viscera**, **lowers** the **intrathoracic pressure**, and **pulls** the **IVC** open. These **actions** facilitate **blood flow** from the **abdomen** into the **thorax**. Slender **branches** of the **right phrenic nerve** and some **lymph vessels** from the **liver** also **traverse** the **foramen** with the **IVC**.

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- **Slender branches** of the **right phrenic nerve** and some **lymph vessels** from the **liver** also **traverse** the **foramen** with the **IVC**.
- **Oesophageal hiatus**: The **oesophagus** pass **obliquely** through an **oval opening**, or a **hiatus**, in the **muscular right crus** of the **diaphragm**, **posterior** to the **central tendon**, **2-3 cm** to the **left** of the **median plane**. It **lies approximately** at the **level** of the **tenth thoracic (T10) vertebra**. Other **structures** passing through the **oesophageal hiatus** are the **anterior** and **posterior vagal trunks**, the **oesophageal branches** of the **left gastric artery**, and **communications** between the **veins** of the **stomach** and **thoracic oesophagus**.

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- The **aortic hiatus: It** lies **posterior** to the **median arcuate ligament** of the **diaphragm** and **anterior** to the **twelfth thoracic vertebra**. The **hiatus** also transmits the **thoracic duct**, the **azygos vein**, and **lymph vessels** which **descend** to the **cisterna chyli** from the **thorax**.
- In addition to **structures** passing through **these three apertures**, a number of **smaller structures** pass between the **thorax** and the **abdomen** through, or around, the **diaphragm**. They are:
- The **phrenic nerve**: the **left phrenic nerve** pierces the **diaphragm lateral** to the **pericardium**. (the **right phrenic nerve** enters the **abdomen** lateral to the **IVC**).


tendon
attachment
Caval hiatus
Oesophageal hiatus
Aortic hiatus
Vertebral

Cont'd

- The **superior epigastric vessels** enter the **abdomen** between the **sternal** and **costal origins** of the **diaphragm.**
- The **musculophrenic vessels** enter the **abdomen** between the slips of **origin** from the **seventh** and **eighth costal cartilages.**
- The **lower five intercostal nerves** enter the abdomen between the slips of **origin** from the **lower six costal cartilages.**
- The **subcostal vessels** and **nerves** enter the **abdomen posterior** to the **lateral arcuate ligament.**
- The **sympathetic trunks** enter the **abdomen posterior** to the **median arcuate ligament.**
- The **splanchnic nerves** pierce the **right** and **crura** of the **diaphragm.**

Relations

- **Thoracic** (superior) **side: lungs/pleura, pericardium, heart.**

- **Abdominal** (inferior) **side: liver** (on **right**), **stomach, spleen, kidneys, intestines.**

Arterial supply

- **Superior surface**
- The **musculophrenic branch** of **internal thoracic artery**, the **pericardiophrenic artery** (branch of **internal thoracic artery**) which **accompanies** the **phrenic nerve** and the **superior phrenic artery** (**branch** of **thoracic aorta**).
- **Inferior surface**
- The **main arteries** supplying the **diaphragm** are the **inferior phrenic arteries** from the **abdominal aorta**.

Venous drainage

- They accompany the arterial supply
- **Musculophrenic vein**
- **Pericardiophrenic vein**
- **Inferior** and **superior phrenic veins**

Nerve supply

- The **motor supply** to the **diaphragm** is from the **phrenic nerves** which **arise** from the **ventral rami** of the **third**, **fourth**, and **fifth cervical spinal nerves** mainly the **fourth**. They **descend** in the thorax, pierce the **diaphragm**, **branch** on the **inferior surface** of the **diaphragm** and **supply** it.
- The **sensory supply** to the **diaphragm** is through the **phrenic** and **lower intercostal nerves**. The **high level** of **origin** of the **phrenic nerves** is due to the **caudal development** of the **diaphragm** and **viscera** during the **development**.

Lymphatic drainage

- Diaphragmatic lymph node
- Parasternal lymph node
- Posterior mediastinal lymph node

Applied Anatomy

- **Diaphragmatic paralysis**: **Injury** to **phrenic nerve** (**trauma**, **surgery**, and **tumor**).
- **Congenital diaphragmatic hernia**: **Protrusion** of **abdominal contents** into the **thoracic cavity** through an **opening** in the **diaphragm**. It is **due** to **posterolateral development defects** in **diaphragm** (**Bochdalek hernia**). **Abdominal contents** herniate into **thorax**, compressing **lungs-pulmonary hypoplasia**.
- **Hiatal hernia (acquired)**: stomach **herniates** through **esophageal hiatus**. Types: **sliding hernia** (most common): **gastroesophageal junction** slides into **thorax**, **paraoesophageal hernia**: **fundus** herniates **beside oesophagus**.
- **Diaphragmatic rupture**: usually due to **blunt trauma** [**left side** more **commonly affected** (**liver** protects the **right**)].

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- **Referred Pain**: due to **phrenic nerve** (C3-C5) **innervation**. **Pain** referred to **shoulder tip** (especially **left shoulder-kehr's sign**, e.g., in **splenic rupture**), **neck** (C3-C5 dermatomes).
- Hiccups: **involuntary spasms** of the **diaphragm**, often **benign**, but can be due to: **gastric distension**, **irritation** of **diaphragm** or **phrenic nerve**.